



T30 Assembly and Disassembly

A DJI Agras T50 agricultural drone is shown in flight, spraying a green field. The drone is white and black, with a large tank and four rotors. It is positioned in the upper left quadrant of the image. A green diamond-shaped graphic is overlaid on the top left corner, containing the word "CONTENTS" in white capital letters.

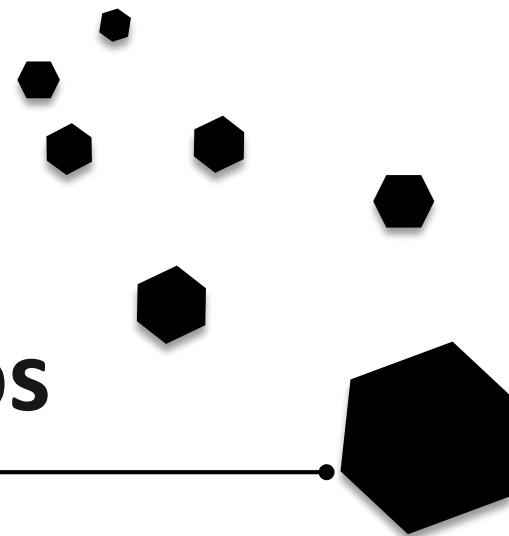
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- 05** / Spreading System Diagram & Notices

01

Part One

Tools and Tutorial Videos



Tools

No.	Tool
1	4.0mm Hexagonal Socket Screwdriver
2	3.0mm Hexagonal Socket Screwdriver
3	2.5mm Hexagonal Socket Screwdriver
4	2.0mm Hexagonal Socket Screwdriver
5	4mm L-type Hexagonal Wrench
6	T15 Torx Screwdriver
7	Long Nose Pliers
8	Cutting Nippers
9	7mm Sleeve
10	8mm Sleeve
11	Anti-Static Tweezers
12	P00 Screwdriver
13	Hose Quick-Release Nut Fixture

No.	Material Name
1	Lubricating Grease
2	7649 Accelerator
3	Liquid PTFE Tape
4	Glue Yellow
5	Loctite 263 Glue
6	LT243 Glue (One Bottle)
7	Loctite_638_Anaerobic Adhesives_Green_275 g/pcs
8	GD-883D White Glue
9	A3 PMU Thermal Gel (300ml/Piece)
10	Debon 4724B UV Adhesive
11	UV Adhesive
12	Black Silicone (K-5911B)
13	Conductive Gel (100ml/stick)
14	Kafuter K-4323N Instant Adhesive
15	Surface Treating Agent (K-4730)

Tutorial Videos

T30 Aircraft Disassembly

<https://cdn.djivideos.com/watch/d535d6e5-54fc-4289-8614-14eb6f71c356>

T30 Aircraft Assembly

<https://cdn.djivideos.com/watch/1b9da2dd-0b42-43ea-85de-3a60f0037c2c>

T30/T10/T20 Remote Controller Disassembly

<https://cdn.djivideos.com/watch/c6379b8d-a0b9-4bdc-8f82-948c97b8844a>

T30/T10/T20 Remote Controller Assembly

<https://cdn.djivideos.com/watch/0f9e9b2f-2ea6-438e-8c81-2bc817832034>

T30/T10 Spreading System 3.0 Spreading System Disassembly

<https://cdn.djivideos.com/watch/598e51de-95bb-4a69-b959-200b3ba450e2>

T30/T10 Spreading System 3.0 Spreading System Assembly

<https://cdn.djivideos.com/watch/44a43826-5f93-4197-8590-aabcce4c24e4>

T30&T10 Charger Disassembly

<https://cdn.djivideos.com/watch/741a94b1-a6b3-4fce-981f-4da0b340f4ce>

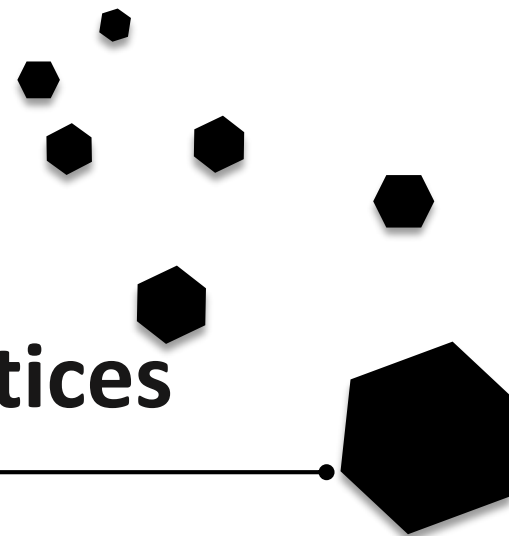
T30&T10 Charger Assembly

<https://cdn.djivideos.com/watch/a5835655-20a6-4c6d-92e5-b4e8acab5a93>

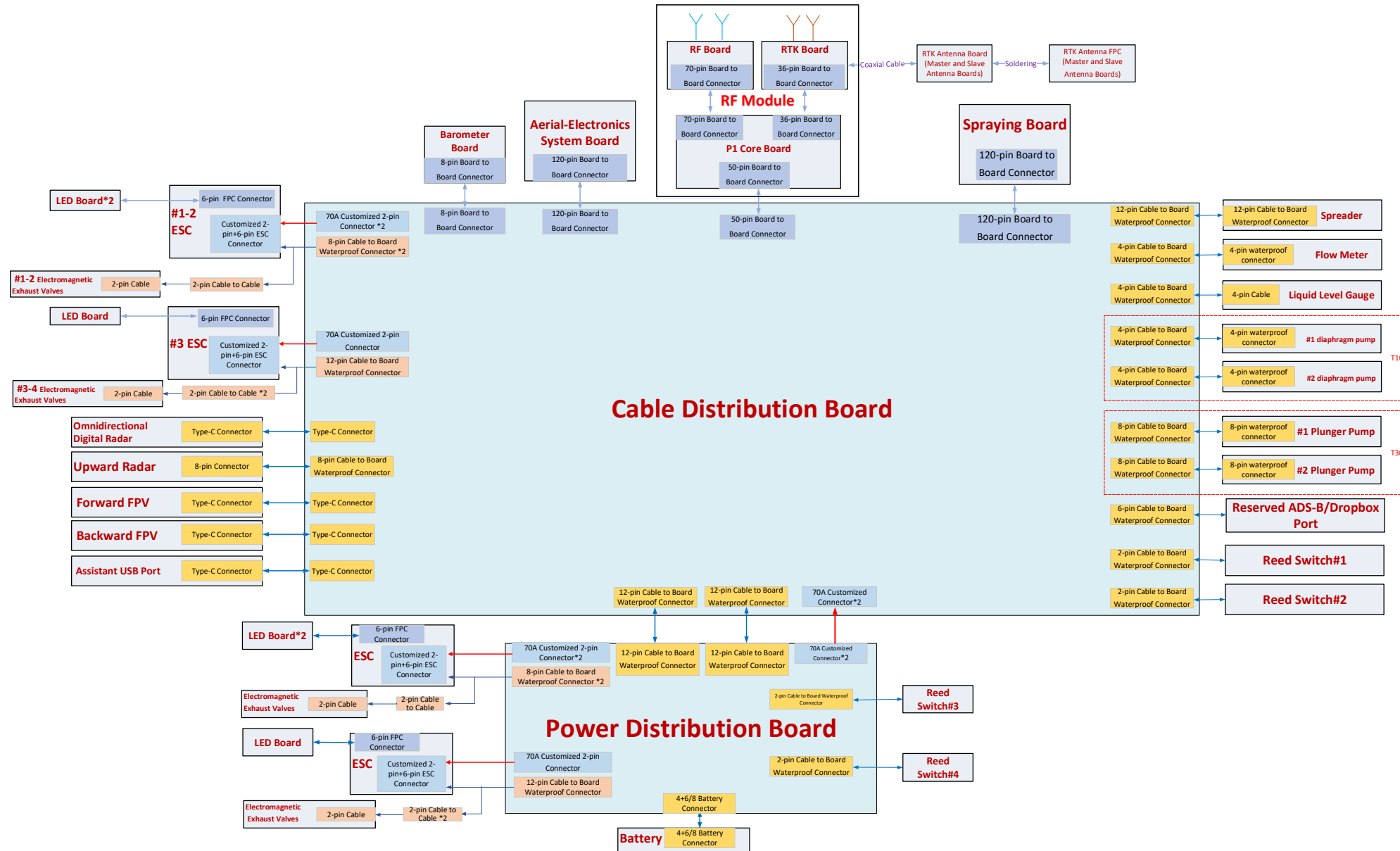
02

Part Two

T30 Aircraft Diagram & Notices



T30 Aircraft Principle Diagram



T30 Aircraft Principle Diagram

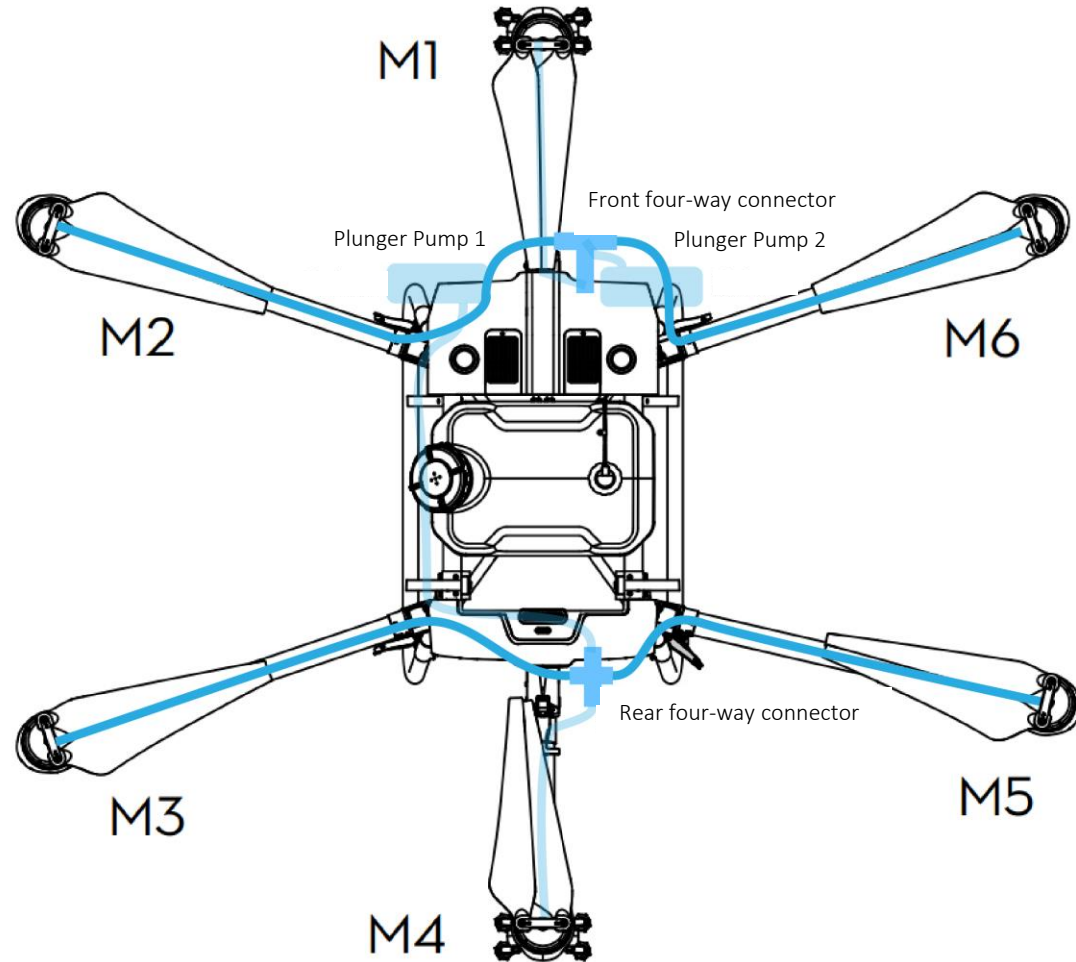
System	Module/Board	Function
Aerial-Electronics System	Aerial-Electronics System Module	The aerial-electronics board is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. The compass is inside the module.
Propulsion System	Motor	The motors work with propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying. The six motors are interchangeable.
	ESC Module	The ESC board converts the throttle information to the motor's rotation speed information, controls the motor rotation, and sends the motor information to the aerial-electronics board.
	LED	The LED board indicates the aircraft's current working condition. Users can know the aircraft's current working condition by the LED's color.
	Power Distribution Board	The power transfer module that distributes the power to the six motors, and is the transfer station for the battery and power supply.
	Cable Distribution Board	Connects the cables and modules and includes a barometer.
	Barometer	Receives the current barometric pressure signal, sends the barometric pressure information to the aerial-electronics board, and detects the aircraft's current altitude information.
Spraying System	Spraying System Module	The control system of the aircraft's core spraying function. It controls the pump and spraying volume.
	Liquid Level Gauge	For T30, it is a continuous liquid level gauge to measure the current liquid amount in the tank to notify the customer whether to add liquid. For T10, similar with T20.
	2-Channel Flow Meter	It is essential for high-precision spraying. It detects the pump's spraying speed. Therefore, the pump's spraying accuracy can be controlled.
	Delivery Pump	The water pump is the exerciser of the aircraft spraying system, which sprays pesticides out by rotating. For T30, the plunger pump is used, and diaphragm pump for T10.
	Electromagnetic Exhaust Valves	Adjust the air in the water pipe, and can independently turn on or off any nozzle to spray.
Positioning System	RF Module (Baseband Board, RF Board, RTK/GPS Board)	As a flight controller system to provide the aircraft position, direction, and speed of the aircraft to ensure the flight stability, and the efficiency and accuracy of the operation. It can track GPS L1/L2, Glonass F1/F2, BDS B1/B2, and Galileo E1/E5b frequency bands at the same time. It has positioning and orientation functions. The GPS module uses the main antenna of RTK module to receive GPS L1, and Glonass F1 frequency band signals.
	Active Antenna Board	Amplify and adjust the weak GPS signal so that the RTK/GPS board can get better satellite signals.
Radar System	Omnidirectional Digital Radar	A sensor for obstacle avoidance and navigation of aircraft. Responsible for investigating the external environment to prevent the aircraft from hurting people and crash.
	Upward Radar	One of the sensors to detect the environment above the aircraft to prevent collisions with obstacles while ascending.
Image Transmission system	Forward and Backward FPV Cameras	Converts optical signals into digital signals, and transmits them to customers through image transmission. The FPV camera contains LEDs to provide night lighting.
	RF Module (Baseband Board, RF Board, RTK/GPS Board)	Includes the baseband board and RF board, processing the image transmission signal and connecting the aircraft and remote controller.
Battery System	Battery	Batteries are the source of energy for flight and spraying. It is an energy storage device.

T30 Aircraft

Spraying System Hose Position

Note:

1. Pay attention to the position of plunger pumps 1 and 2.
2. Pay attention to the connection method between the hoses and two delivery pumps.



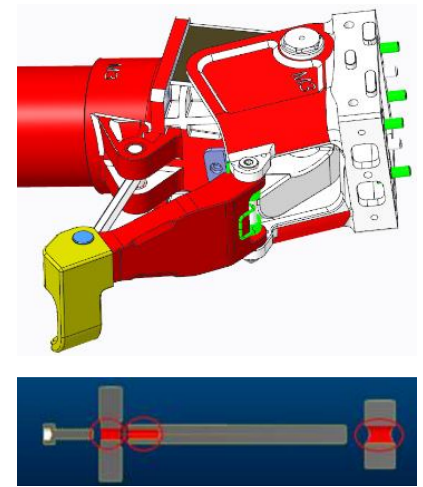
T30 Aircraft Notices



When replacing the propellers or propeller holder, the screw needs to be replaced and screw glue needs to be applied.



When replacing the screw on the motor or replacing the motor, make sure the screw is mounted in place. Clean off the screw glue and then apply more glue again. Otherwise, the motor may become loose and the aircraft may noticeably shake.



When replacing the aircraft arm lock, apply the FL.SC.H00193 Loctite 263 Red Threadlocker to the position as shown on the red dot.

T30 Aircraft Notices



After the arm lock is replaced, fasten the lever as shown. The tension value needs to be tested (acceptable tension range: $90\pm 10\text{N}$).



The landing gear fixing base needs to be mounted facing outwards.

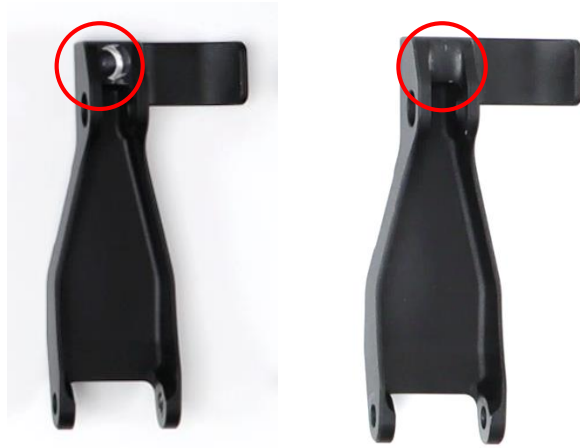
The front left and rear right landing gear fixing bases are interchangeable.

The front right and rear left landing gear fixing bases are interchangeable.

T30 Aircraft Notices



Pay attention to the installation direction of the landing gear, and the radar signal cable needs to be attached to the left side (when the aircraft nose faces forward).



There is no magnet inside the front and rear two aircraft arm levers. Distinguish between the left and right four aircraft arms.

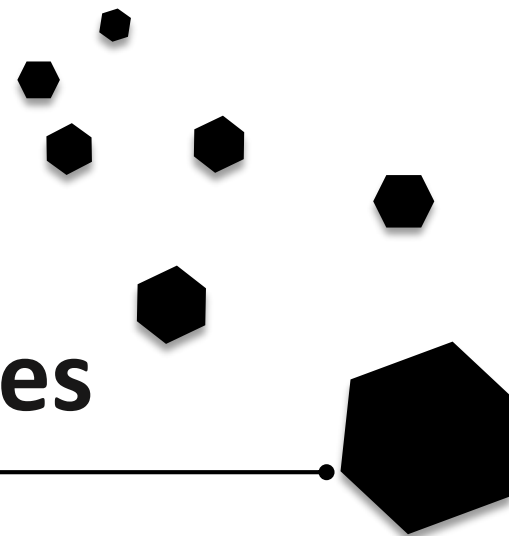


Distinguish between the direction of the left and right middle frame carbon tubes.

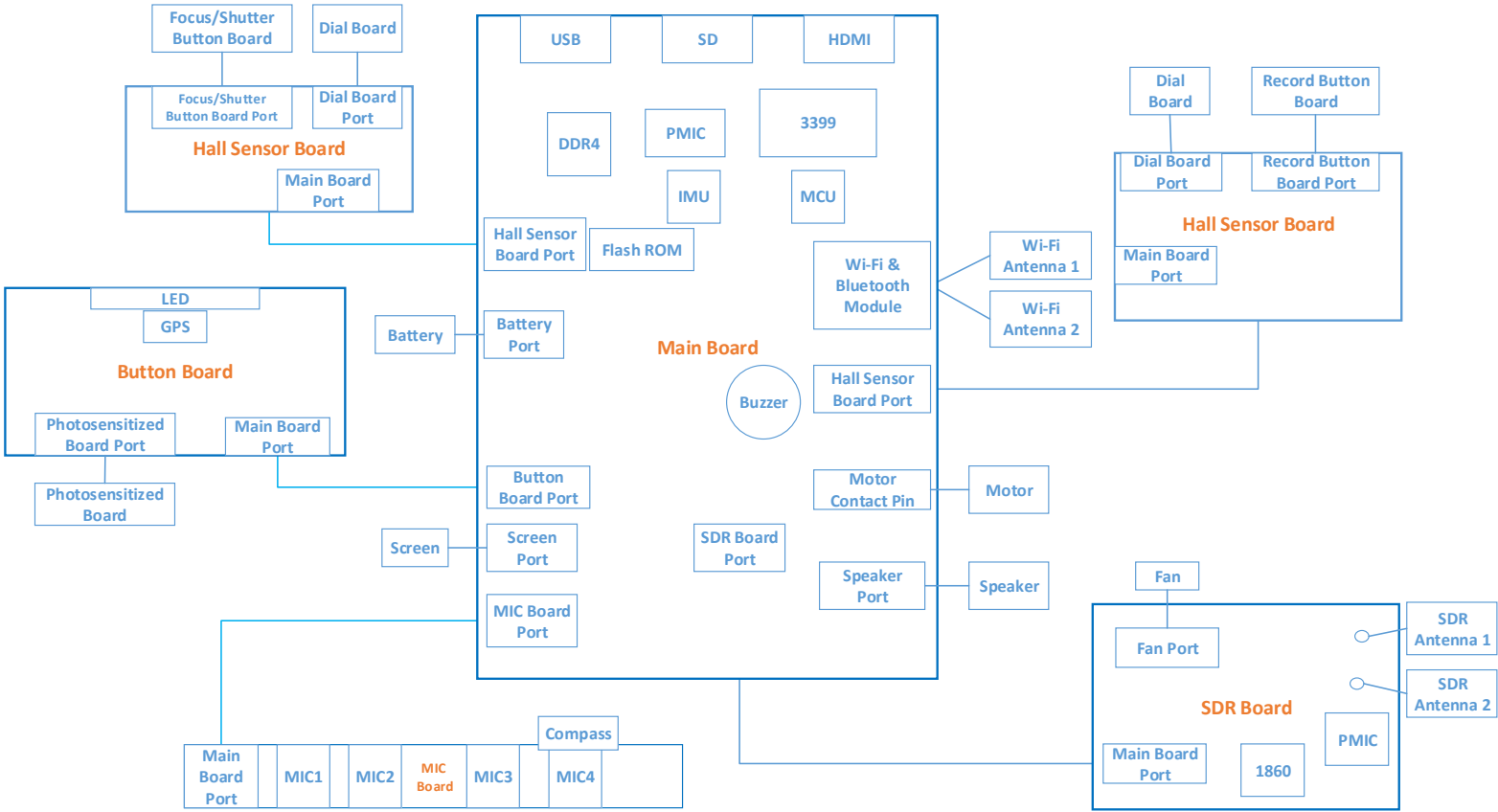
03

Part Three

T30 RC Diagram & Notices



T30 Remote Controller Principle Diagram



Chip	Function
3399	Main functional chip
MCU	Micro control unit, 1549 chip
PMIC	Power management chip
DDR4	Fourth-generation memory chip; temporary storage unit for commands and data.
Flash ROM	Store program and data
IMU	Inertial measurement unit; including accelerometer and gyroscope, related to gravity sensing
1860	Wireless image transmission bandwidth chip; used for digital signal processing, encoding, and digital signal modulation.

T30 Remote Controller Notices



Removing the lower shell

There are buckles between the upper and lower shells, which are very firm after assembly. When removing the lower shell, first unscrew all the screws, hold the lower shell with your hand, and then remove it forcefully.



Removing the front cover's decorative piece

When removing the components inside the front shell, remove the front cover's decorative piece first. Remove the two screws, and then remove the front cover's decorative piece by using the crowbar to pry it off gently.

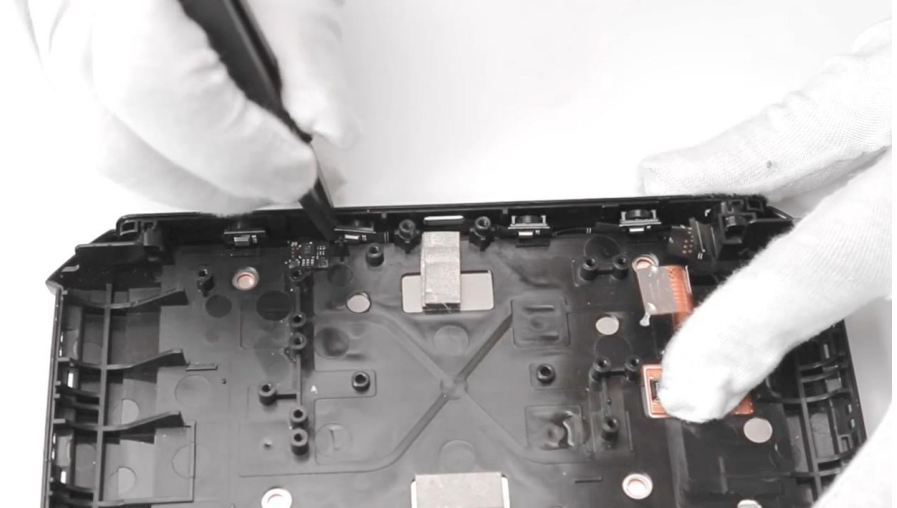
T30 Remote Controller Notices



Removing the antenna sleeve

When replacing the image transmission antenna, keep in mind that glue was applied during assembly to make the product very firm, so the heat gun needs to first be used to blow on the antenna sleeve (minimum temperature and minimum air volume). When the sleeve becomes misshapen, shake the sleeve to the left and right, and then remove and discard it.

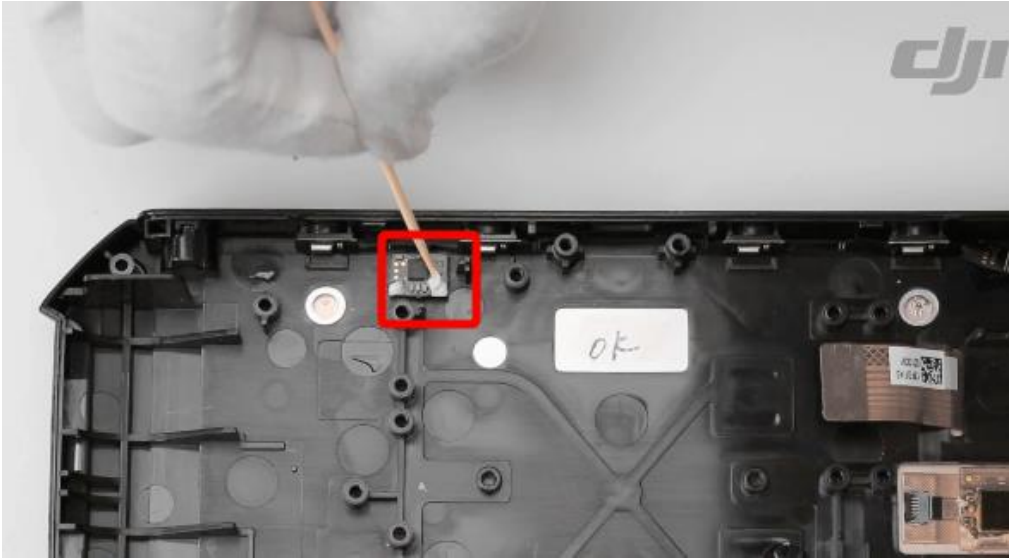
Do not damage the antenna base with the heat gun.



Removing the Mic board

There are four fixing pieces (buckles) on the Mic board. Observe the mounting position, and then use the tweezers to eject them.

T30 Remote Controller Notices



Mounting the Mic board

When the Mic board is mounted, a hot-melt process is used. After repair, white glue needs to be applied.



Mounting left and right soft rubber

Apply the Kafuter instant adhesive before mounting the left and right pieces of soft rubber.

Mounting the 5D button: When the 5D button is replaced, the Kafuter instant adhesive needs to be applied. Then apply a small amount of hot melt glue to the button on the button board.

T30 Remote Controller Notices



Soft rubber



Front cover's decorative piece



Antenna sleeve

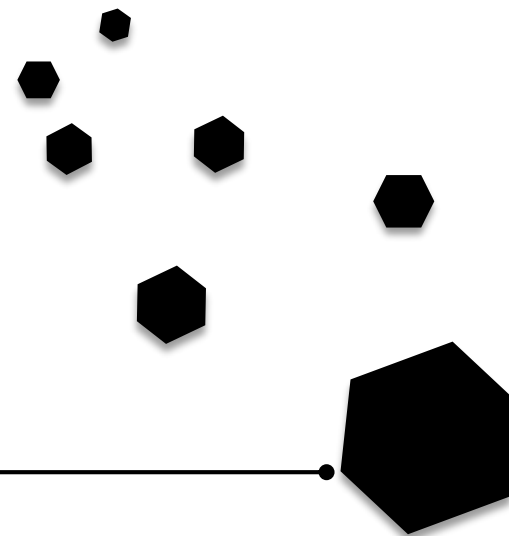
Consumption Materials

Soft rubbers, Front cover's decorative piece, Antenna sleeves, these 3 materials are consumption materials, replace a new one when disassembling these parts.

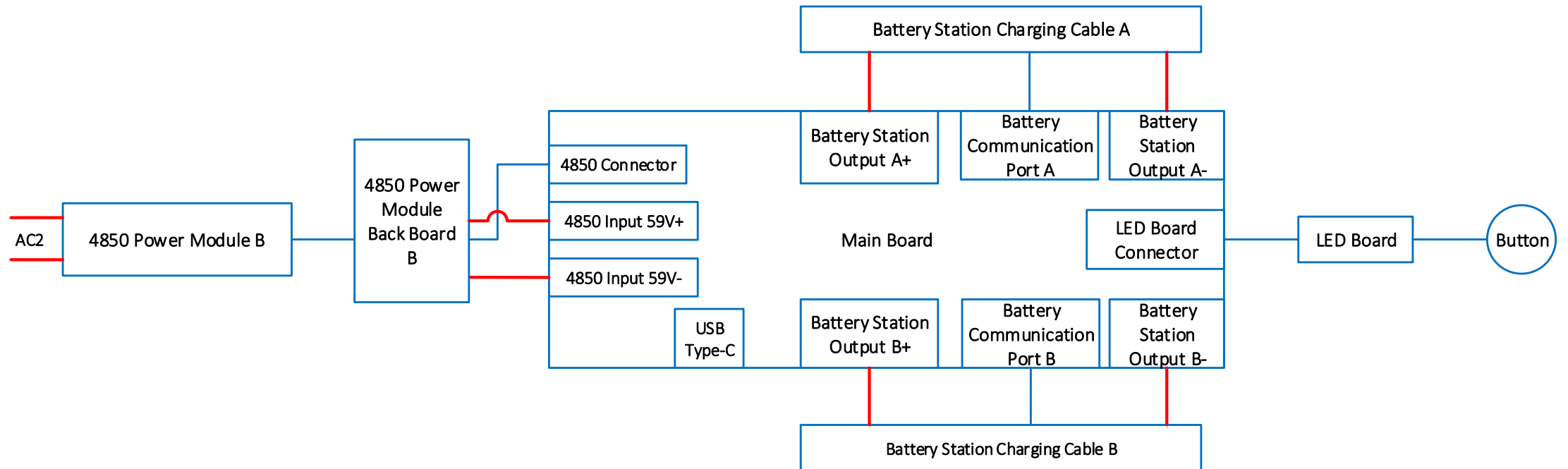
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Part Four

T30 Battery Station Diagram & Notices



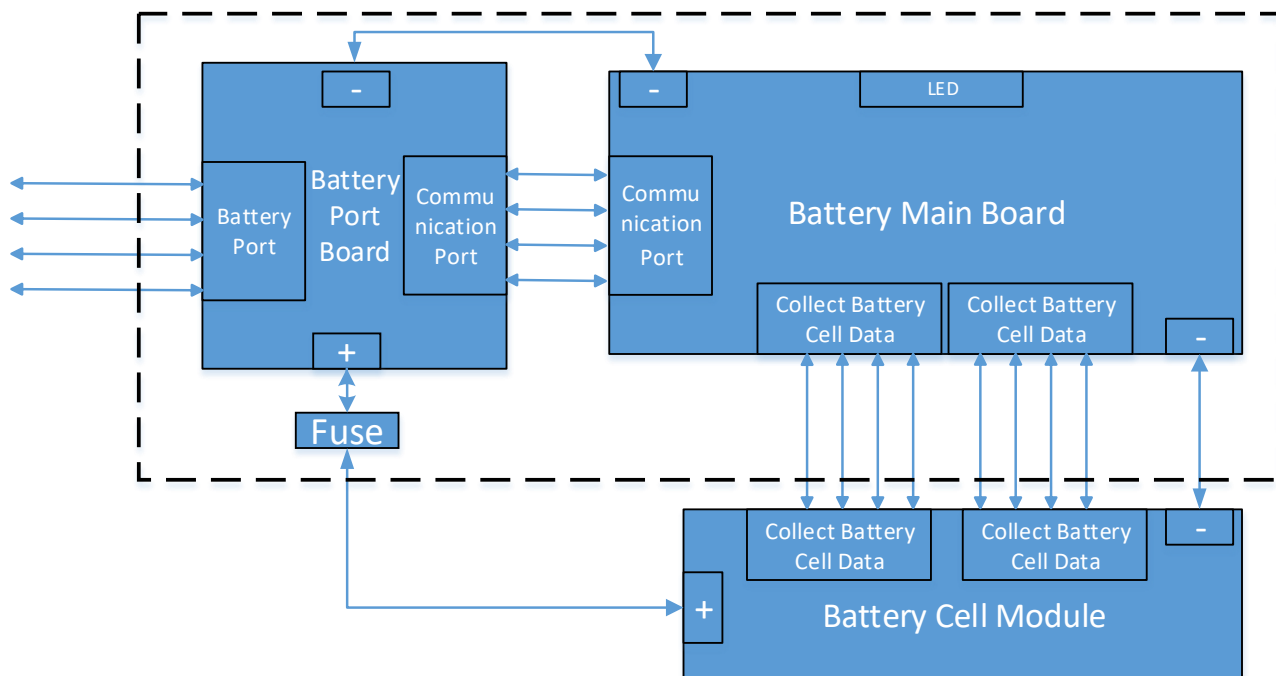
T30 Battery Station Principles Diagram



T30 Battery Station Principles Diagram

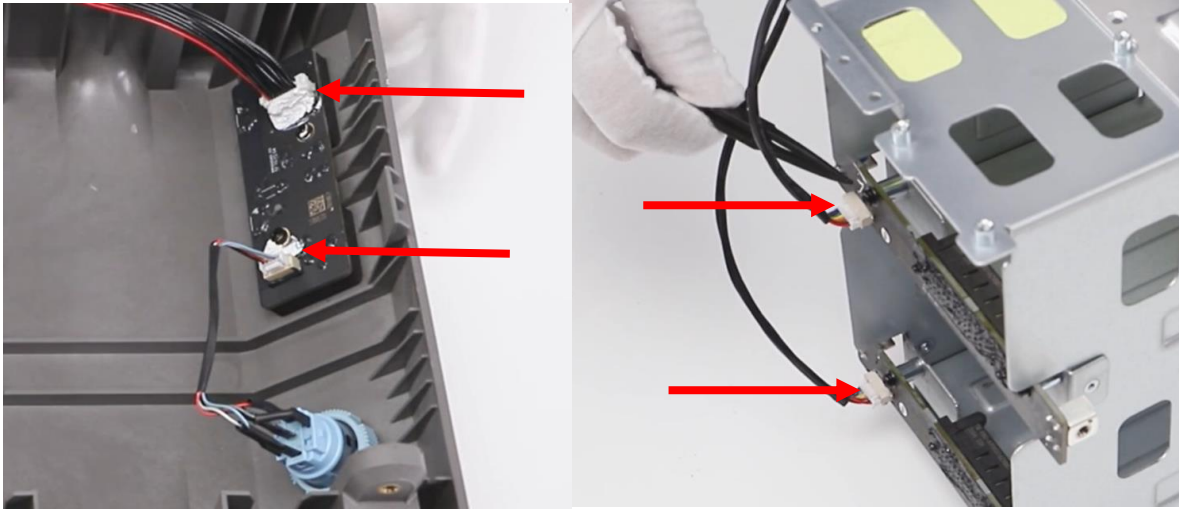
Chip	Function
4850 Power Module	The 220V electric power is converted into 59V direct current to charge the battery; also communicates with the main board. The maximum output current of a single power module can reach 65A.
4850 Module Power Back Board	Transfers power between the 4850 module and the main board. The power output of the power module is transferred to the main board through the 4850 module. The 4850 module communicate with the main board through the back board.
Battery Station Main Board	Communicates with the power module and battery, manages the battery charging current, and outputs the LED signal to control the LED display. The management board firmware can be updated by connecting to the computer via Type-C port or on the remote controller. The log can be exported through the computer and imported via the remote controller.
Output Cable	An adaptive cable between the main board and battery, providing charging voltage and communication.
LED Board	Connects the management board and the system switch respectively, and transmits the switch signal to the management board through the LED board. The LED on the LED board is used to display the battery charging status and the status of the battery station.

T30 Battery Principles Diagram



Chip	Function
Battery Cell Module	The core part of the battery and the source of battery power supply and charging. Main parameters: Number of cycles, internal resistance, voltage difference, which are used to judge the quality of the battery cell. Provided by two suppliers: ALT (number of cycles: 1000) and Desay (number of cycles: 1000)
Fuse	When the interface is short-circuited or abnormal, the current is too large and the fuse will be burnt out to protect the battery.
Battery Main Board	Communicates with the battery station or aircraft, and manages and calculates the capacity of the battery cell module. The LED can display the battery status.
Battery Adaptive Board	Connects the battery cell, and connects the battery main board to the external charger and aircraft.

T30 Battery Station Notices

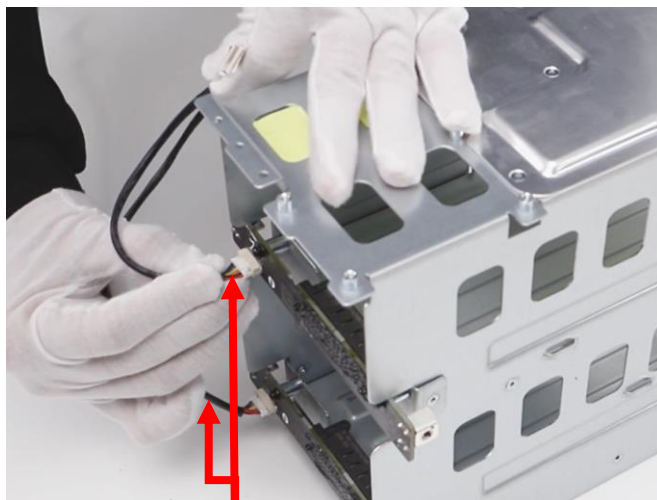


All flat **cable connectors** should be applied the white glue.



The **Power Button** is No fool-proof design; make sure the power icon faces upward.

T30 Battery Station Notices

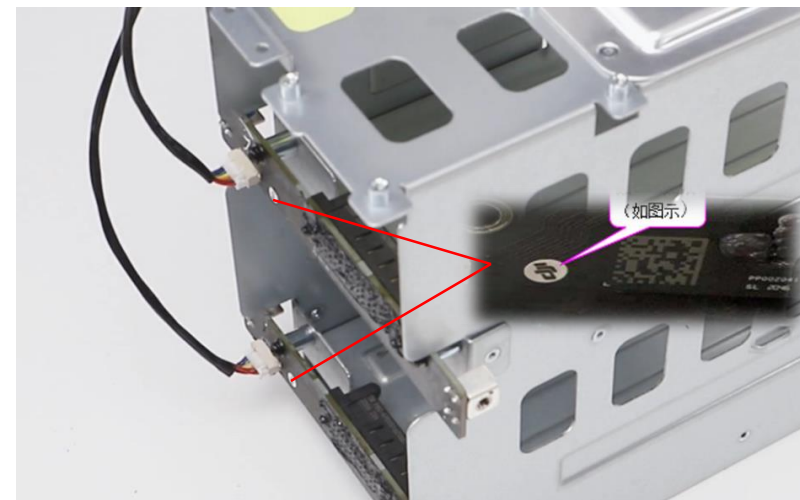


Communication cables



U-Type frame

The **communication cables** needs to be attached first, and then mount the bracket module to the **U-type frame**.

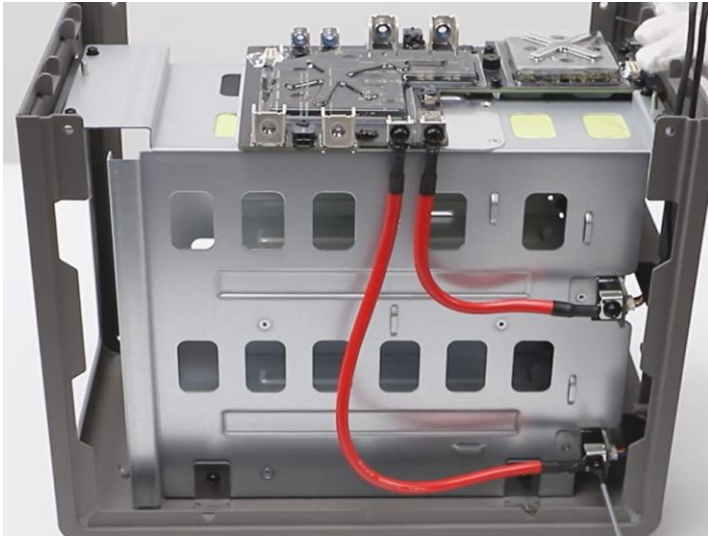


Waterproof Label

If there is no waterproof label on a received power module adapter board, attach a waterproof label if the port board needs to be replaced.

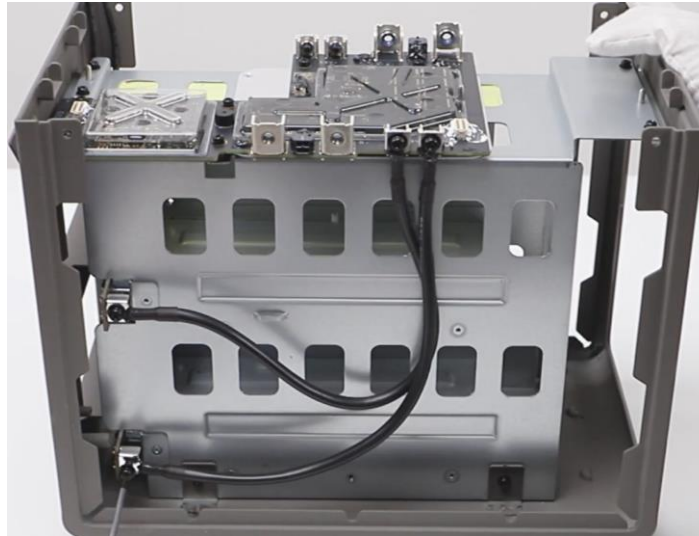
T30 Battery Station Notices

The torque for the cables



Positive Cables

$10.0 \pm 0.5 \text{ kgf.m}$



Negative Cables

$8.0 \pm 0.5 \text{ kgf.m}$



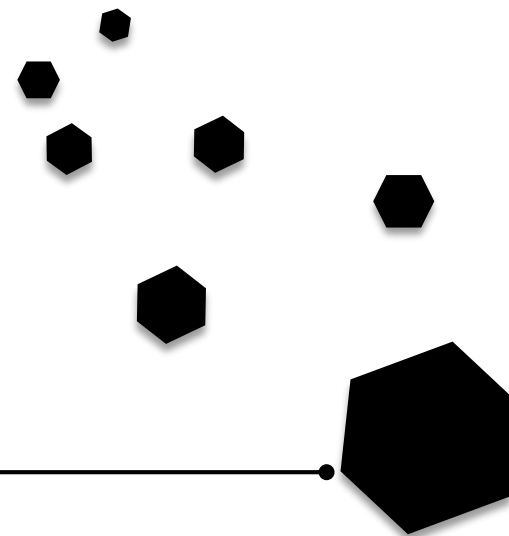
Charging Cables

$8.0 \pm 0.5 \text{ kgf.m}$

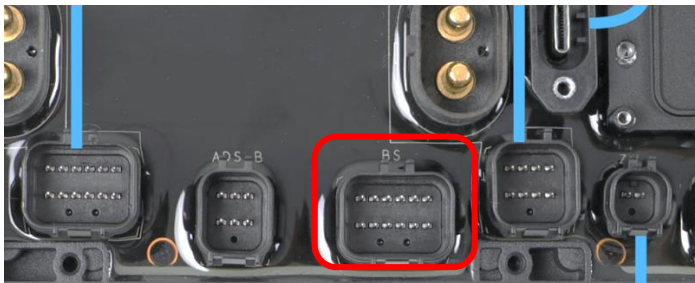
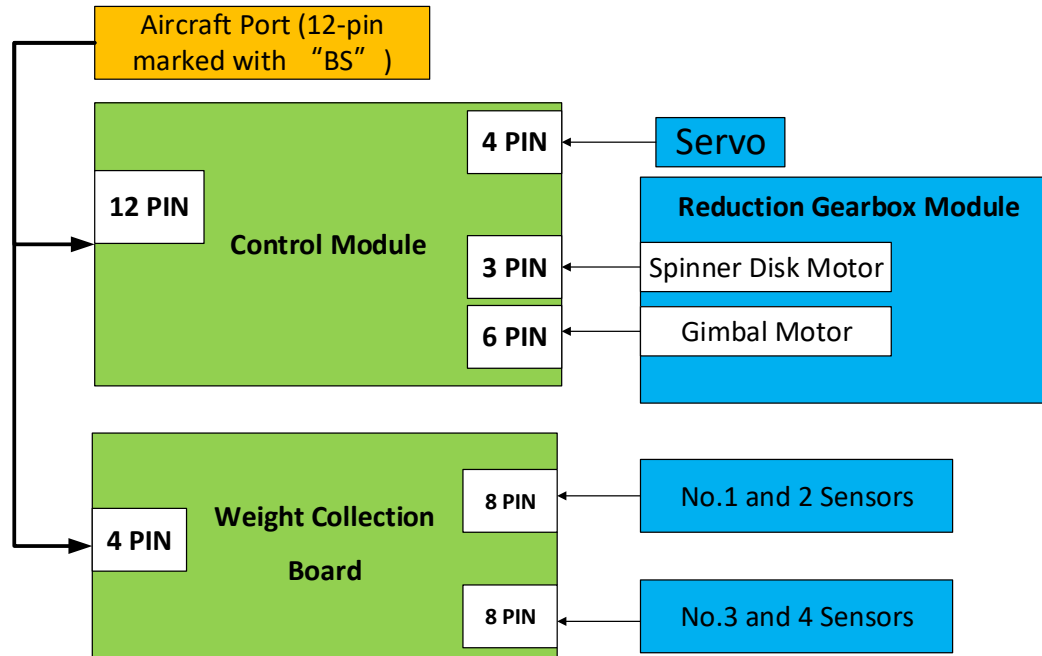
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Part Five

T30 Spreading System Diagram & Notices

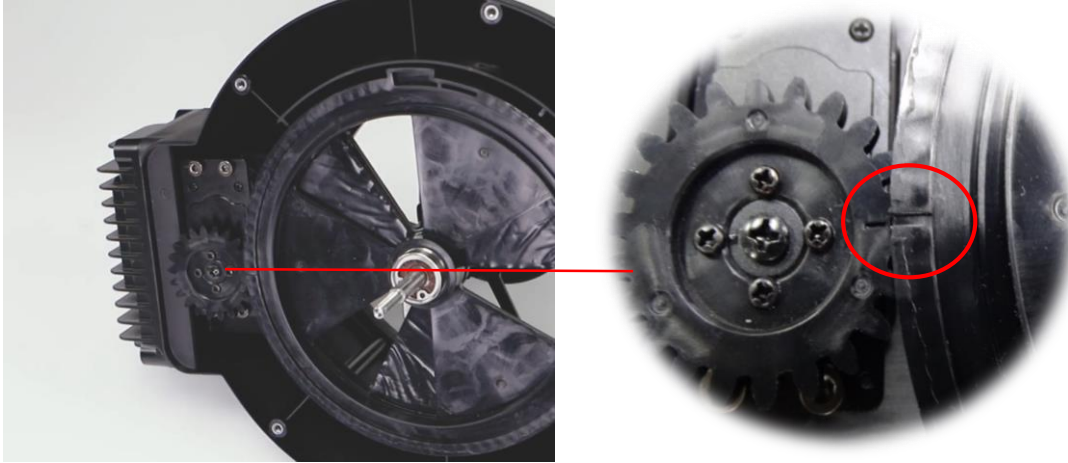


T30 Spreading System Principles Diagram



Chip	Function
Control Module	Core module of the spreader; collects Hall sensor signal, ESC signals, and servo signals, and controls the various resources of the entire spreader.
Weight Collection Board	Receives the data from the weight sensor and calculate the weight, then transmits the data to the control module for app prompts of an empty tank.
Weight Sensor	Collects the weight of the tank.
Servo	Controls the change of the hopper gate angle and the opening/closing of the hopper gate.
Reduction Gearbox Module	There are two built-in motors. The gimbal motor drives the structure attached above the motor to rotate. The control module determine whether to make an app prompt for an empty tank based on the current value of the motor; the spinner disk motor drives the spinner to rotate and spreads the materials that fall on the spinner disk.

T30 Spreading System Notices



When installing the large gear, make sure that the large gear is aligned with the notch of the small gear on the servo module.



On the weight control module, the number of the weight sensor modules on the left side is 1 and 2, and that of weight sensor modules on the right side is 3 and 4.

The first three digits of the serial number for No.1 to No.4 weight sensor modules are 4JT, 4QG, 4QH, 4QJ respectively.

THANK YOU

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